



Effect of pressure and the capability of 2-methoxyethanol as a solvent in the behaviour of a diisopropyl ether–isopropyl alcohol azeotropic mixture

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Abstract

Consistent vapour–liquid equilibrium data for the binary and ternary systems of diisopropyl ether+isopropyl alcohol at 30 and 101.3 kPa and diisopropyl ether+2-methoxyethanol, isopropyl alcohol+2-methoxyethanol and diisopropyl ether+isopropyl alcohol+2-methoxyethanol at 101.3 kPa are reported. The activity coefficients of the solutions were correlated with its composition using the Wilson, NRTL and UNIQUAC models. It is shown that the models allow a very good prediction of the phase equilibrium of the ternary system using the pertinent parameters of the binary systems. Moreover, the effect of pressure and the entrainer capability of 2-methoxyethanol were studied.

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1. Introduction

The most common method of manufacturing isopropyl alcohol is by the dehydration of propylene using sulphuric acid as the catalyst [1]. However, before the isopropyl alcohol can be removed from the reaction mixture, some of it reacts with sulphuric acid to form diisopropyl ether. Isopropyl alcohol and diisopropyl ether form a binary homogeneous azeotrope. Normally a mixture of several solvents are separated and recovered by fractionation in a rectification column and the ease of separation depends upon the difference in boiling points of the compounds to be separated. However, the separation of a homogeneous azeotrope mixture, that is a common task in the chemical industry, cannot be carried out by conventional distillation. For the solution of the separation problem, there are several techniques, including extractive distillation [2] and pressure-swing distillation [3]. Extractive dis-

tillation can be used to separate the components of an azeotropic mixture by adding an agent (entrainer) that modifies the relative volatility of the mixture although a previous selection of a possible entrainer is required. In this work, the selection has been done by Scheibel's criterion [2], and by means of the calculation of the separation factor. We chose 2-methoxyethanol (widely known by its trade name, methyl cellosolve) which is a common solvent in laboratory practice and in the chemical industry. Also, pressure-swing distillation (PSD) is often mentioned as an alternative process to the broad applied processes azeotropic distillation and extractive distillation. In PSD the dependency of the azeotropic concentration on the system pressure is used to overcome the azeotrope. The general advantages of the PSD process are the saving of energy and the abdication of entrainers compared to well-known alternatives.

The thermodynamic analysis, prediction and computer simulation of phase equilibrium help to understand and evaluate the different separation processes. In any case, a reliable knowledge of the phase equilibrium behaviour is necessary for synthesis, design and optimisation of any distillation system.

In this work, to improve the knowledge of the phase behaviour we report VLE data for the binary system diisopropyl ether (1)+isopropyl alcohol (2) at two pressures (30 and 101.3 kPa). Moreover, we measured isobaric VLE data for

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the ternary system diisopropyl ether (1)+isopropyl alcohol (2)+2-methoxyethanol (3) and two constituent binary systems diisopropyl ether (1)+2-methoxyethanol (3) and isopropyl alcohol (2)+2-methoxyethanol (3) at 101.3 kPa. In a recent literature review only isobaric VLE for the diisopropyl ether (1)+isopropyl alcohol (2) system at atmospheric pressure has been found [4].

VLE data of binary and ternary systems were found to be thermodynamically consistent. Data reduction was carried out using the Wilson, NRTL and UNIQUAC equations to relate activity coefficients with compositions.

2. Experimental

2.1. Chemicals

Diisopropyl ether (>99.0 mass%, analytical grade) was purchased from Fluka, isopropyl alcohol (>99.5 mass%, HPLC grade) and 2-methoxyethanol (>99.8 mass%, analytical grade) were purchased from Aldrich Ltd. The reagents were used without further purification after chromatography failed to show any significant impurities. The water content, determined using a Karl Fischer volumetric automatic titrator (Metrohm, 701 KF Titrino), was small in all chemicals (<0.05 mass%). Before measurements, the liquids were degassed and subsequently dried over molecular sieves (Union Carbide, type 4 Å, 1/16 in. pellets). The refractive indexes of the pure components were measured at 298.15 K using an Abbe refractometer Atago 3T, and the densities were measured at 298.15 K using an Anton Paar DMA 58 densimeter. Temperature was controlled to ± 0.01 K with a thermostated bath. The accuracy in refractive index and density measurements are ± 0.0002 and $\pm 0.01 \text{ kg m}^{-3}$, respectively. The experimental values of these properties and the boiling points are given in Table 1 together with those given in the literature. Appropriate precautions were taken when handling the reagents in order to avoid hydration.

2.2. Apparatus and procedure

The equilibrium vessel used in the measurements (Labodest VLE 602/D) was an all-glass dynamic-recirculating still equipped with a Cottrell circulation pump, manufactured by Fischer Labor und Verfahrenstechnik (Germany). The Cottrell pump ensures that both liquid and vapour phases are in intimate contact during boiling and also in contact with the temperature sen-

sing element. The equilibrium temperature was measured with a digital Hart Scientific thermometer model 1502A and a Pt 100 probe Hart Scientific model 5622 calibrated at the ENAC-accredited Spanish *Instituto Nacional de Técnica Aeroespacial*. The accuracy is estimated to be ± 0.01 K. The temperature probe was checked against the ice and steam points of distilled water. A Fisher M101 pressure control system was used to measure and control the pressure and the heating power. The pressure accuracy is ± 0.1 kPa. The manometer was calibrated using the vapour pressure of ultrapure water.

In each experiment, the pressure was fixed and the heating and stirring system of the liquid mixture was turned on. The still was operated at constant pressure until equilibrium was reached. Equilibrium conditions were assumed when constant temperature and pressure were obtained for 30 min or longer. Then, samples of liquid and condensate were taken for analysis. The sampling was carried out with special syringes that allowed withdrawal of small volume samples.

2.3. Analysis

The composition of the sampled liquid and condensed vapour phase were determined using a CE Instruments GC 8000 Top gas chromatograph (GC), after calibration with gravimetrically prepared standard solutions. A flame ionization detector was used together with a 30 m, 0.454 mm i.d., capillary column (DB-MTBE, J & Scientific). The GC response peaks were treated with Chrom-Card for Windows. Column, injector and detector temperatures were 363, 473, and 498 K, respectively, for all systems. Very good peak separation was achieved under these conditions and calibration analyses were carried out to convert the peak area ratio to the mass composition of the sample. In general, two analyses were made of each liquid and vapour composition. The standard deviation in the mole fraction was usually less than 0.001.

3. Results and discussion

3.1. Pure component vapour pressures

The pure component vapour pressure for 2-methoxyethanol was determined experimentally as a function of the temperature using the same equipment as that for obtaining the VLE data. The pertinent results appear in Table 2. The measured vapour

Table 1
Density d , refractive index n_D , and normal boiling point T_b of pure components

Component	$d \text{ (kg m}^{-3}\text{) (298.15 K)}$		$n_D \text{ (298.15 K)}$		$T_b \text{ (K)}$	
	Exptl.	Lit.	Exptl.	Lit.	Exptl.	Lit.
Diisopropyl ether (1)	0.7183	0.7182 ^a	1.3652	1.3655 ^a	341.49	341.45 ^a
Isopropyl alcohol (2)	0.7827	0.7813 ^a	1.3754	1.3752 ^a	355.35	355.41 ^a
2-Methoxyethanol (3)	0.9601	0.9602 ^a	1.4002	1.4023 ^b	397.44	397.55 ^b

^a TRC tables [5].

^b Thornton and Garner [6].

Table 2
Vapour pressure of 2-methoxyethanol

<i>T</i> (K)	<i>P</i> (kPa)
336.18	9.93
340.22	11.93
343.74	13.91
346.79	15.85
349.63	17.88
352.13	19.87
354.46	21.84
357.64	24.83
360.51	27.82
363.12	30.80
365.53	33.78
367.73	36.71
369.84	39.72
371.79	42.67
374.27	46.68
376.57	50.68
378.60	54.44
381.20	59.57
383.54	64.51
385.74	69.50
387.80	74.44
389.80	79.45
391.65	84.39
393.36	89.12
395.15	94.33
397.24	100.76
397.44	101.30

pressures were correlated using the Antoine equation:

$$\ln P_i^0 \text{ (kPa)} = A_i - \frac{B_i}{T \text{ (K)} + C_i} \quad (1)$$

whose parameters A_i , B_i and C_i are reported in Table 3 together with some literature values.

The deviations, $\Delta P_i^0 = P_{i,\text{lit.}}^0 - P_{i,\text{exptl.}}^0$, calculated by means of the Antoine equation using the constant values from Table 3 have been graphically represented in Fig. 1. In this figure, it can be observed that the mean error was less than 0.5% with respect to the values reported by Daubert and Danner [7] and less than 1.0% with respect to the data reported by Pick et al. [8].

The pure vapour pressures of diisopropyl ether and isopropyl alcohol were taken from Daubert and Danner [7].

Table 3
Vapour pressure parameters

Compound	Eq. ^a	A_i	B_i	C_i	D_i	E_i	Reference
Diisopropyl ether (1)	(2)	41.631	−4668.70	−12.792	6.3693×10^{-4}	1	Daubert and Danner [7]
Isopropyl alcohol (2)	(2)	92.935	−8177.10	−10.031	3.9988×10^{-6}	2	Daubert and Danner [7]
2-Methoxyethanol (3)	(1)	15.05	3499.67	−62.061			This work
	(2)	202.63	−12472.00	−27.385		2	Daubert and Danner [7]
	(3)	7.95	1793.98	236.88	2.6400×10^{-5}		Pick et al. [8]

^a Vapour pressure equations: (1) $\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} + C]$; (2) $\ln P^0 \text{ (Pa)} = A + B/T \text{ (K)} + C \ln T \text{ (K)} + D(T \text{ (K)})^E$; (3) $\log P^0 \text{ (mmHg)} = A - B/[T \text{ (°C)} + C]$.

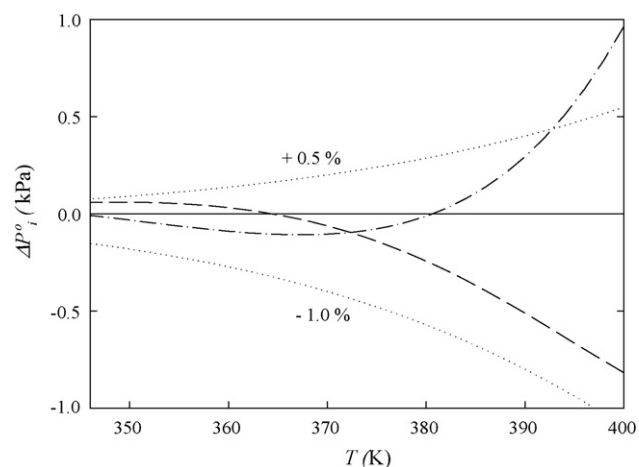


Fig. 1. Differences, $\Delta P_i^0 = P_{i,\text{lit.}}^0 - P_{i,\text{exptl.}}^0$, obtained for vapour pressures of the 2-methoxyethanol using the parameters given in Table 3. (—) Daubert and Danner [7]; (---) Pick et al. [8].

3.2. Binary systems

The temperature T , the liquid phase x_i and vapour phase y_i mole fractions for the system diisopropyl ether (1) + isopropyl alcohol (2) at 30 and 101.3 kPa and diisopropyl ether (1) + 2-methoxyethanol (3), diisopropyl alcohol (2) + 2-methoxyethanol (3) systems at 101.3 kPa are reported in Tables 4–6 and plotted in Figs. 2–4. The activity coefficients γ_i for these systems were calculated from the following equation:

$$\gamma_i = \frac{y_i P}{x_i P_i^0} \quad (2)$$

where y_i is the mole fraction of component i in vapour phase, x_i the mole fraction of component i in liquid phase, γ_i the activity coefficient of component i in liquid phase, P (kPa) the total pressure and P_i^0 (kPa) is the saturation vapour pressure for the pure liquid i . In Eq. (2), the vapour phase is assumed to be an ideal gas and the pressure dependence of the liquid phase fugacity is neglected. Eq. (2) was selected to calculate the activity coefficients as the data measured at low pressure which makes the simplification in the equation reasonable.

The activity coefficients presented in Tables 4–6 indicate that the diisopropyl ether (1) + isopropyl alcohol (2) and diisopropyl alcohol (1) + 2-methoxyethanol (3) systems shows positive deviations from ideal behaviour and isopropyl alcohol (2) + 2-methoxyethanol (3) system exhibits no deviation from ideal behaviour and no additional azeotropes are present.

Table 4

Experimental vapour–liquid equilibrium data for the binary system diisopropyl ether (1) + isopropyl alcohol (2) at 30 and 101.3 kPa

<i>P</i> (kPa)	<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₂
30.0	327.84	0.000	0.000		1.000
	324.48	0.025	0.121	2.517	1.066
	320.62	0.076	0.317	2.533	1.066
	317.83	0.127	0.430	2.283	1.090
	315.41	0.186	0.535	2.132	1.084
	313.47	0.247	0.606	1.953	1.106
	312.10	0.309	0.653	1.775	1.145
	310.82	0.382	0.693	1.602	1.218
	309.98	0.447	0.729	1.490	1.259
	309.31	0.508	0.754	1.394	1.334
	308.75	0.566	0.775	1.314	1.430
	308.32	0.626	0.794	1.240	1.555
	307.89	0.684	0.813	1.182	1.709
	307.60	0.746	0.830	1.119	1.968
	307.37	0.796	0.851	1.086	2.179
	307.20	0.847	0.873	1.054	2.496
	307.18	0.898	0.901	1.027	2.922
	307.25	0.941	0.933	1.011	3.436
	307.47	0.977	0.969	1.003	4.041
	307.72	1.000	1.000	1.000	
101.3	355.35	0.000	0.000		1.000
	352.60	0.029	0.104	2.549	1.024
	350.39	0.066	0.210	2.427	1.028
	348.69	0.102	0.290	2.291	1.032
	347.12	0.142	0.361	2.141	1.040
	345.58	0.189	0.427	1.996	1.053
	344.46	0.232	0.475	1.871	1.070
	343.38	0.281	0.517	1.739	1.103
	342.50	0.331	0.557	1.634	1.130
	341.75	0.382	0.586	1.528	1.180
	341.07	0.439	0.619	1.435	1.234
	340.46	0.499	0.649	1.347	1.311
	340.05	0.555	0.675	1.277	1.393
	339.73	0.613	0.700	1.214	1.494
	339.47	0.672	0.727	1.160	1.621
	339.31	0.728	0.755	1.117	1.770
	339.28	0.785	0.784	1.075	1.987
	339.40	0.842	0.820	1.045	2.233
	339.71	0.894	0.861	1.024	2.532
	340.37	0.947	0.920	1.011	2.844
	341.49	1.000	1.000	1.000	

For the diisopropyl ether (1) + isopropyl alcohol (2) binary systems, minimum boiling azeotropes are present at 30 and 101.3 kPa. The compositions and boiling temperatures of the azeotropes are summarised in Table 7, together with the corresponding literature values. In all cases, the azeotropic composition was obtained from the best polynomial fit of the function ($y_1 - x_1$) = $f(x_1)$. Azeotropic temperatures were obtained from the best polynomial fit for $T = f(x_1)$, using the x_1 values previously determined. In Table 7, one can observe the strong dependency of azeotropic compositions on pressure.

The test of Fredenslund et al. [11] was applied to the binary experimental data to test thermodynamic consistency. In Table 8 the parameters of the Legendre polynomial together with the pertinent statistics required by the Fredenslund test are given. The residuals for all systems at 101.3 kPa show a reasonable random distribution.

Table 5

Experimental vapour–liquid equilibrium data for the binary system diisopropyl ether (1) + 2-methoxyethanol (3) at 101.3 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₃
397.44	0.000	0.000		1.000
393.20	0.009	0.124	3.470	1.011
387.75	0.022	0.279	3.452	1.010
382.46	0.038	0.410	3.413	1.003
377.01	0.057	0.527	3.343	0.994
371.72	0.079	0.620	3.231	0.993
365.79	0.115	0.711	3.008	0.981
359.47	0.168	0.788	2.726	0.983
356.41	0.211	0.822	2.470	0.982
354.28	0.258	0.844	2.207	1.002
352.30	0.313	0.860	1.968	1.056
350.78	0.369	0.871	1.770	1.124
349.41	0.427	0.880	1.612	1.224
348.43	0.480	0.887	1.489	1.330
347.58	0.534	0.895	1.386	1.429
346.74	0.593	0.899	1.289	1.621
346.01	0.650	0.905	1.212	1.829
345.21	0.710	0.914	1.148	2.079
344.45	0.779	0.925	1.084	2.474
343.75	0.834	0.935	1.047	2.918
343.03	0.887	0.949	1.022	3.524
342.24	0.941	0.971	1.011	3.962
341.49	1.000	1.000	1.000	

The activity coefficients were correlated with the Wilson [12], NRTL [13], and UNIQUAC [14] equations. The parameters of these equations were obtained by minimizing the following objective function (OF):

$$\text{OF} = \sum_{i=1}^N \left(\left| \frac{T_i^{\text{exptl.}} - T_i^{\text{calc.}}}{T_i^{\text{exptl.}}} \right| + \left| y_i^{\text{exptl.}} - y_i^{\text{calc.}} \right| \right) \quad (3)$$

Table 6

Experimental vapour–liquid equilibrium data for the binary system isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa

<i>T</i> (K)	<i>x</i> ₂	<i>y</i> ₂	<i>γ</i> ₂	<i>γ</i> ₃
397.44	0.000	0.000		1.000
392.64	0.050	0.186	1.029	0.997
388.36	0.099	0.328	1.046	0.997
384.14	0.155	0.446	1.040	1.009
380.64	0.207	0.540	1.057	1.006
377.42	0.261	0.622	1.076	0.992
374.67	0.314	0.686	1.083	0.979
372.03	0.371	0.738	1.081	0.980
369.77	0.425	0.775	1.074	1.000
367.62	0.485	0.812	1.065	1.011
365.79	0.541	0.842	1.058	1.021
363.99	0.600	0.875	1.059	0.993
362.33	0.658	0.900	1.057	0.991
360.84	0.715	0.921	1.053	0.996
359.48	0.767	0.937	1.052	1.025
358.64	0.815	0.950	1.036	1.059
357.96	0.859	0.963	1.023	1.056
357.33	0.902	0.975	1.011	1.053
356.59	0.942	0.985	1.007	1.100
355.35	1.000	1.000	1.000	

Table 7
Azeotropic compositions and temperatures

System	<i>P</i> (kPa)	<i>T</i> (K)		<i>x</i> ₁	
		Exptl.	Lit.	Exptl.	Lit.
Diisopropyl ether (1) + isopropyl alcohol (2)	30	307.25	—	0.9064	—
	101.3	339.31	339.35 ^a 339.25 ^b	0.7818	0.7870 ^a 0.8000 ^b

^a Verhoeve [9].

^b Yorzane et al. [10].

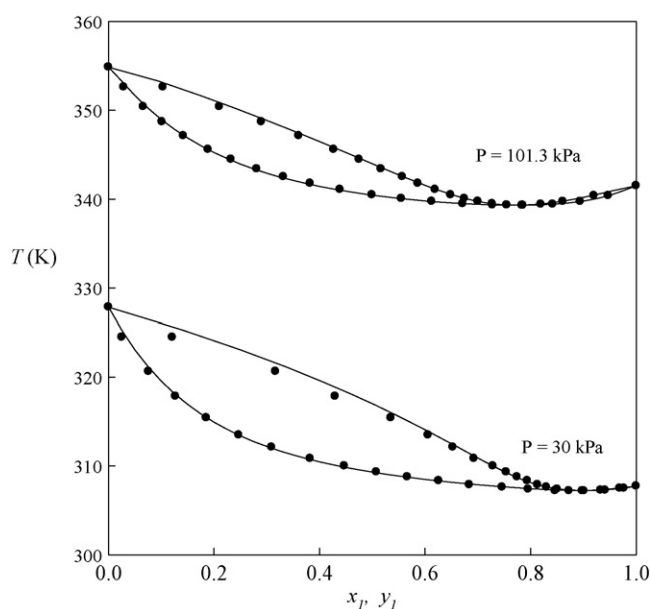


Fig. 2. Experimental data for the system diisopropyl ether (1) and isopropyl alcohol (2) at 30 and 101.3 kPa: (●) experimental data; (—) calculated using Wilson model with the parameters given in Table 9.

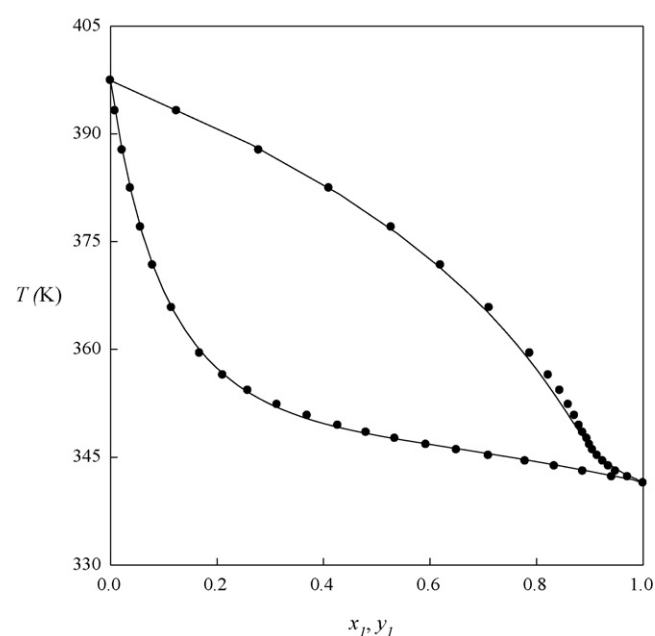


Fig. 3. Experimental VLE data for the system diisopropyl ether (1) + 2-methoxyethanol (3) at 101.3 kPa: (●) experimental data; (—) smoothed data using the Wilson model with the parameters given in Table 9.

and are reported in Table 9, together with the pertinent statistics of each VLE correlation.

Moreover, for the diisopropyl ether (1) + isopropyl alcohol (2) system an additional correlation of experimental data at 30 and 101.3 kPa was carried out for the three models taking into account the dependence of the excess energy on temperature. The parameters for these models were considered to be linear function of the temperature.

$$A_{ij} = A_{ij}^0 + A_{ij}^1 T \quad (4)$$

Table 8
Consistency test for the binary systems with diisopropyl ether (1), isopropyl alcohol (2) and 2-methoxyethanol (3)

System <i>i</i> + <i>j</i>	<i>P</i> (kPa)	<i>A</i> ₁ ^a	<i>A</i> ₂ ^a	<i>A</i> ₃ ^a	AAD <i>y</i> _{<i>i</i>} ^b	AAD <i>P</i> ^c (kPa)
1–2	30	1.2985	0.1342	0.1189	0.0087	0.07
	101.3	1.1413	0.1211	0.0546	0.0064	0.27
1–3	101.3	1.4123	0.0806	0.0066	0.0064	0.67
2–3	101.3	0.1539	0.1618	0.0400	0.0028	0.35

^a Legendre polynomial parameters.

^b Average absolute deviation in vapour phase composition.

^c Average absolute deviation in pressure.

The results given in Table 10 show that all models fitted well the experimental data of this system.

3.3. Ternary system

VLE data for the ternary system diisopropyl ether (1) + isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa are reported in Table 11. The ternary data were found to be thermodynamically consistent by the Wisniak and Tamir [16] modification of the McDermott–Ellis [17] test ($D < D_{\max}$ at all data points) and the Wisniak *L–W* test [18] ($0.92 < L_i/W_i < 1.10$).

Table 9
Parameters and deviations between experimental and calculated values for different G^E models for the systems with diisopropyl ether (1), isopropyl alcohol (2) and 2-methoxyethanol (3) at 101.3 kPa

Model	System $i + j$	A_{ij} (J mol ⁻¹)	A_{ji} (J mol ⁻¹)	α_{ij}	Bubble point		
					AAD T^a	AAD y_1^b	AAD y_2^b
Wilson ^c	1 + 2	-476.15	4397.34		0.07	0.0064	
	1 + 3	695.40	4608.79		0.17	0.0078	
	2 + 3	-2007.52	4140.21		0.14		0.0036
	1 + 2 + 3 ^d				0.48	0.0084	0.0049
NRTL	1 + 2	2733.36	898.07	0.30	0.11	0.0066	
	1 + 3	2543.68	2241.59	0.30	0.16	0.0058	
	2 + 3	299.58	142.80	0.30	0.45		0.0067
	1 + 2 + 3 ^d				0.48	0.0076	0.0046
UNIQUAC ^e	1 + 2	2561.80	-877.48		0.11	0.0071	
	1 + 3	2076.82	-416.02		0.18	0.0059	
	2 + 3	-1027.58	1493.35		0.59		0.0094
	1 + 2 + 3 ^d				0.45	0.0088	0.0053

^a Average absolute deviation in temperature.

^b Average absolute deviation in vapour phase composition.

^c Molar liquid volumes of pure components from DECHEMA [15].

^d Ternary prediction from binary parameters.

^e Volume and surface parameters from DECHEMA [15].

VLE data for the ternary system have been predicted using the Wilson, NRTL and UNIQUAC equations with the binary interaction parameters obtained from the regression of binary data. Table 9 lists the mean absolute deviations between experimental and calculated temperature and vapour phase mole fractions. From these results, it can be concluded that the binary contributions allow a good prediction of the ternary system, representing the data successfully. Thus, the models can be used to calculate boiling points from liquid phase compositions at the system pressure. Fig. 5 shows a comparison of VLE data experimental and the data calculated with the UNIQUAC model with the parameters given in Table 9.

3.4. Study of the azeotropic composition with the pressure

In Table 7, the azeotropic data obtained in this work for both pressures are given together with the literature data [9,10]. The results obtained for diisopropyl ether (1) + isopropyl alcohol (2) system show that the azeotropic point is very sensitive to pressure and the mole fraction of diisopropyl ether at this point decreases with the pressure.

In Fig. 6 the VLE curve at 10 and 400 kPa simulated by Aspen Split v2006 AspenTech Ltd. [19] using the Wilson model with the experimental parameters (Table 10) together with the experimental data (30 and 101.3 kPa) are shown. As can be seen in this figure, the VLE curve at 10 kPa is very close to the diagonal (relative volatility close to 1.0) at high diisopropyl ether concentration, although in this case the binary azeotrope is eliminated. Nevertheless, working at 10 kPa is not economically competitive with other azeotropic-breaking techniques (azeotropic or extractive distillation). On the other hand, working at 400 kPa, the shift in azeotropic composition due to the change of the pressure is large enough and it can be thought about a process where two distillation columns operate at the two different pressures 101.3 and 400 kPa. For this reason, it is thought that pressure-swing distillation takes advantage against the other alternative techniques.

3.5. Influence of the solvent in the azeotropic mixture

In order to carry out the separation of the azeotropic mixture of diisopropyl ether (1) + isopropyl alcohol (2) by extractive

Table 10
Global parameters and deviations between experimental and calculated values for different G^E Models for the diisopropyl ether (1) + isopropyl alcohol (2) system at 30 and 101.3 kPa

Model	A_{12}^0 (J mol ⁻¹)	A_{12}^1 (J mol ⁻¹ K ⁻¹)	A_{21}^0 (J mol ⁻¹)	A_{21}^1 (J mol ⁻¹ K ⁻¹)	α_{ij}	Bubble point	
						AAD T^a	AAD y_1^b
Wilson ^c	-1549.91	2.48	4638.95	0.80		0.26	0.0084
NRTL	2781.35	2.15	1044.94	-2.22	0.30	0.23	0.0066
UNIQUAC ^d	2966.70	-0.96	-787.93	-0.38		0.17	0.0069

^a Average absolute deviation in temperature.

^b Average absolute deviation in vapour phase composition.

^c Liquid volumes from DECHEMA [15].

^d Volume and surface parameters from DECHEMA [15].

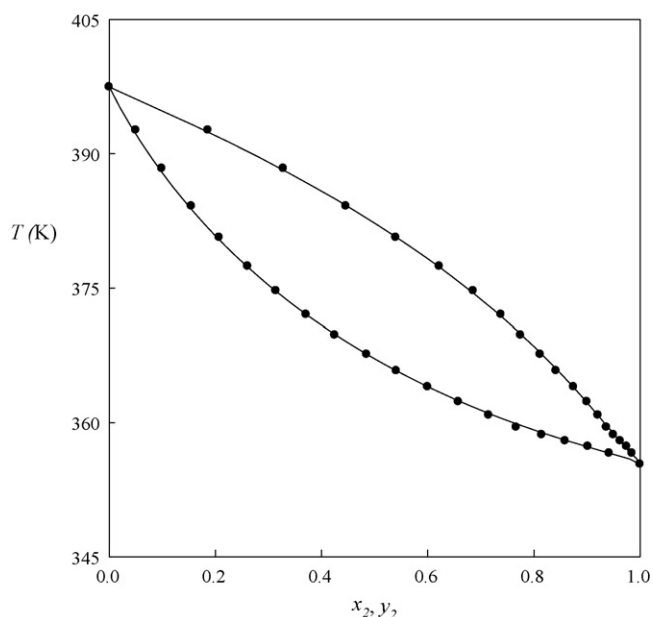


Fig. 4. Experimental VLE data for the system isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa: (●) experimental data; (—) smoothed data using the Wilson model with the parameters given in Table 9.

distillation, we studied the influence of 2-methoxyethanol as a solvent on the phase equilibria behaviour of that azeotropic mixture by means of a topological analysis. The residue curve map [20] and relative volatility analysis [21] are included in the main recent publications in chemical engineering as important tools for distillation design of azeotropic and extractive trains of columns. In Fig. 7 residue curves, at 101.3 kPa, simulated by Aspen Split v2600 using the UNIQUAC model with the experimental parameters (Table 9) are shown. As can be seen in this figure, all residue curves approaching the 2-methoxyethanol (solvent) vertex are inflected toward the 2-methoxyethanol–isopropyl alcohol face, with the result that

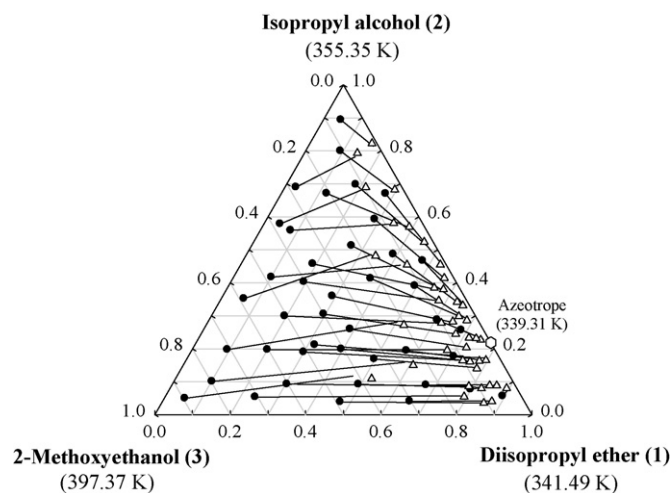


Fig. 5. Comparison of vapour phase composition for the system diisopropyl ether (1) + isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa: (●) experimental liquid phase mole fraction; (△) vapour phase mole fraction; (—) calculated with the UNIQUAC equation with the parameters given in Table 9.

Table 11

Experimental vapour–liquid equilibrium data for diisopropyl ether (1) + isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>x</i> ₂	<i>y</i> ₁	<i>y</i> ₂	<i>γ</i> ₁	<i>γ</i> ₂	<i>γ</i> ₃
340.05	0.684	0.257	0.750	0.227	1.154	1.687	3.356
340.83	0.894	0.057	0.893	0.080	1.024	2.612	4.677
341.01	0.605	0.287	0.736	0.231	1.241	1.474	2.517
341.07	0.477	0.467	0.651	0.331	1.389	1.294	2.698
341.25	0.704	0.177	0.795	0.165	1.143	1.688	2.772
341.73	0.494	0.391	0.684	0.286	1.379	1.299	2.110
341.88	0.798	0.077	0.862	0.088	1.071	2.011	3.233
342.74	0.389	0.486	0.631	0.342	1.564	1.194	1.719
343.35	0.570	0.194	0.779	0.163	1.292	1.387	1.884
343.48	0.674	0.090	0.843	0.089	1.178	1.621	2.173
343.73	0.318	0.212	0.716	0.233	2.103	1.792	0.803
343.76	0.277	0.670	0.530	0.455	1.784	1.104	2.169
344.30	0.287	0.592	0.560	0.414	1.792	1.111	1.539
344.56	0.366	0.413	0.657	0.298	1.634	1.133	1.461
344.92	0.655	0.041	0.873	0.041	1.199	1.538	2.010
345.22	0.497	0.169	0.783	0.140	1.404	1.266	1.598
346.28	0.388	0.259	0.724	0.204	1.608	1.146	1.366
346.41	0.265	0.513	0.575	0.380	1.861	1.078	1.339
346.58	0.494	0.092	0.826	0.080	1.429	1.250	1.496
346.87	0.184	0.698	0.453	0.523	2.078	1.070	1.323
346.98	0.395	0.199	0.755	0.161	1.613	1.145	1.336
347.69	0.291	0.360	0.649	0.282	1.838	1.082	1.237
348.01	0.472	0.039	0.854	0.035	1.477	1.239	1.393
348.32	0.295	0.305	0.676	0.245	1.851	1.075	1.218
349.89	0.299	0.190	0.734	0.165	1.889	1.090	1.130
350.45	0.190	0.457	0.549	0.385	2.185	1.034	1.055
350.52	0.092	0.799	0.295	0.682	2.415	1.046	1.176
350.88	0.120	0.671	0.390	0.569	2.422	1.026	1.062
351.15	0.304	0.093	0.790	0.086	1.927	1.109	1.113
351.18	0.194	0.403	0.580	0.346	2.215	1.024	0.991
352.34	0.047	0.894	0.164	0.822	2.522	1.048	1.141
352.71	0.195	0.299	0.621	0.278	2.252	1.042	1.013
353.88	0.200	0.197	0.677	0.195	2.311	1.061	1.017
354.00	0.239	0.054	0.793	0.055	2.253	1.085	1.035
356.30	0.081	0.559	0.343	0.581	2.693	1.011	0.924
357.49	0.101	0.417	0.441	0.454	2.687	1.014	0.896
358.80	0.030	0.690	0.141	0.793	2.805	1.016	0.932
360.08	0.043	0.579	0.214	0.690	2.828	1.003	0.948
362.84	0.094	0.197	0.524	0.272	2.949	1.044	0.965
363.62	0.060	0.352	0.344	0.482	2.935	1.007	0.964
364.02	0.101	0.101	0.610	0.150	3.069	1.080	0.963
373.80	0.055	0.050	0.520	0.109	3.704	1.118	0.919

isopropyl alcohol and 2-methoxyethanol will be recovered in the bottom and diisopropyl ether in the distillate, as can be expected from the binary systems behaviour.

Thermodynamic topological analysis of the relative volatility provides, also, a very useful tool for the screening of entrainers. Relative volatility in the presence of the solvent is a valuable tool demonstrating the effect of the solvent on the binary system diisopropyl ether (1) + isopropyl alcohol (2) and is defined as follows:

$$\alpha_{12}^S = \frac{y_1/x_1}{y_2/x_2} = \frac{\gamma_1 P_1^0}{\gamma_2 P_2^0} \quad (5)$$

In Fig. 7, the isovolatility line ($\alpha_{12}^S = 1$) is presented for the binary system diisopropyl ether (1) + isopropyl alcohol (2) in presence of 2-methoxyethanol. Laroche et al. [21] demonstra-

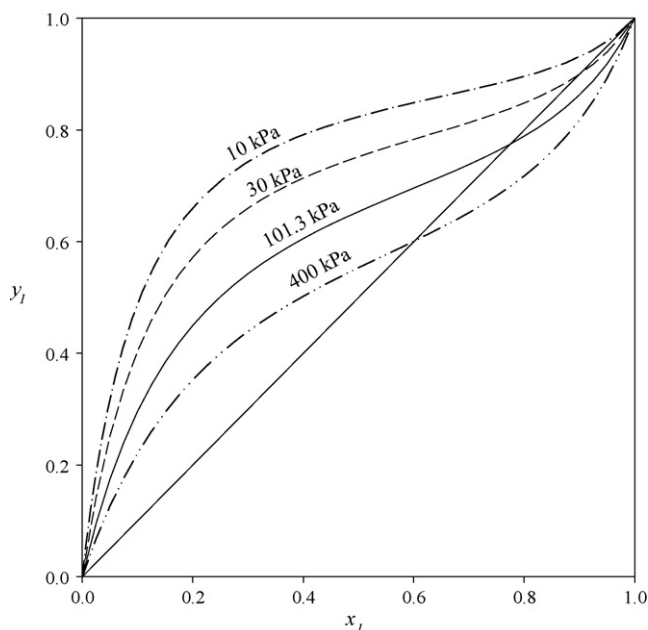


Fig. 6. Estimated composition diagram for the diisopropyl ether (1)–isopropyl alcohol (2) system at different pressures using the UNIQUAC model with the parameters given in Table 10.

ted for several ternary mixtures that if these isovolatility lines intersect the 1–3 (solvent) edge, component 1 is recovered as distillate in an extractive column while component 2 is recovered as distillate in the entrainer recovery column. So, diisopropyl ether will be recovered in the distillate using 2-methoxyethanol as entrainer. Moreover, the requirements for the solvent to be used in an extractive column are related to the position of the intersection of the isovolatility curve on the binary 1–3 side. As can be seen in Fig. 7, the position of the isovolatility line $\alpha_{12}^S = 1$ with respect to 2-methoxyethanol + diisopropyl ether mixture makes necessary relatively low concentration of 2-methoxyethanol (at least 0.24 in molar fraction of 2-

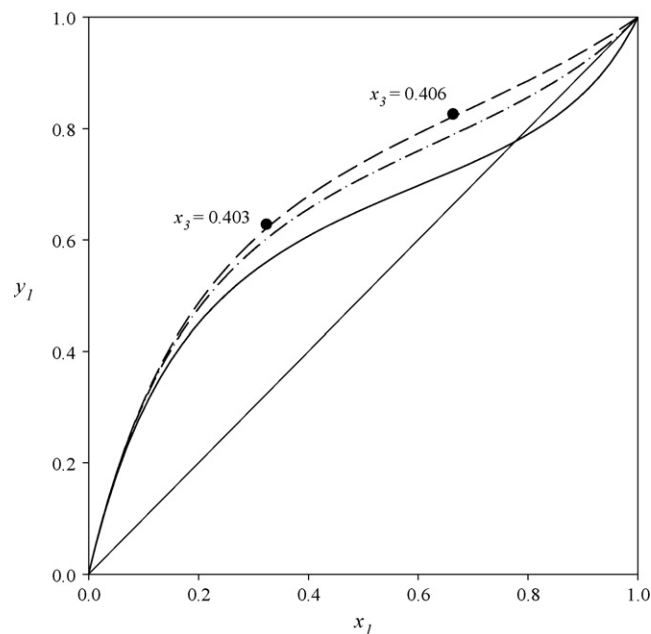


Fig. 8. VLE data plotted on a solvent free basis for the system diisopropyl ether (1) + isopropyl alcohol (2) + 2-methoxyethanol (3) at 101.3 kPa. Experimental data: (—) for $x_3 = 0.00$; (●) experimental points. Calculated using the UNIQUAC equation with the parameters given in Table 9: (---) for $x_3 = 0.24$; (---) for $x_3 = 0.40$.

methoxyethanol in the azeotropic feed plate) in the extractive column.

Another useful alternative is the study of the solvent influence on the equilibrium behaviour of the azeotropic mixture, on a solvent free basis. As can be observed in Fig. 8, 2-methoxyethanol (3) eliminates the diisopropyl alcohol (1)–isopropyl alcohol (2) azeotrope, allowing the separation of pure diisopropyl ether from isopropyl alcohol by rectification when they are employed as solvents in extractive distillation. 2-Methoxyethanol enhances the relative volatility of diisopropyl ether to isopropyl alcohol ($\alpha_{12}^S = 2.46$).

4. Conclusions

Consistent VLE data have been determined for the binary systems diisopropyl ether + isopropyl alcohol at 30 and 101.3 kPa and diisopropyl ether + 2-methoxyethanol and isopropyl alcohol + 2-methoxyethanol and the ternary system diisopropyl ether + isopropyl alcohol + 2-methoxyethanol at 101.3 kPa. The Wilson, NRTL and UNIQUAC models were capable of correlating all the binary systems and yielded reasonable predictions for the ternary system.

According to the equilibrium diagrams for the binary system diisopropyl ether + isopropyl alcohol, it is shown that the azeotropic point is very sensitive to pressure. Therefore, in view of the results, pressure-swing distillation could be a useful technique in order to overcome the binary azeotrope.

The ternary experimental data are analyzed on the thermodynamic topological maps and the 2-methoxyethanol free basis. It is shown that 2-methoxyethanol is an excellent solvent to

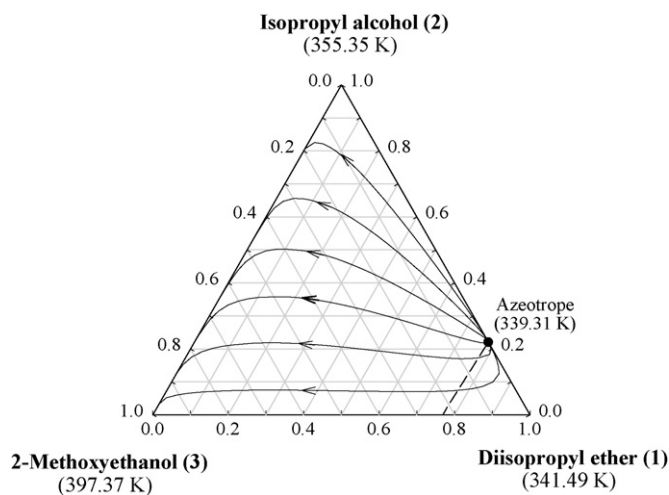


Fig. 7. Residue curve map and isovolatility line for the ternary diisopropyl ether (1) + isopropyl alcohol (2) + 2-methoxyethanol (3): (—) residue curves simulated by Aspen Split using UNIQUAC model with the parameters given in Table 9; (---) isovolatility line; (●) azeotrope.

break the azeotrope in the mixture (diisopropyl ether + isopropyl alcohol).

List of symbols

A, B, C, D, E vapour pressure equation parameters

P work pressure (kPa)

P^0 pure component vapour pressure

R universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)

T absolute temperature (K)

x, y compositions of the liquid and vapour phases

Greek letters

α_{ij} NRTL parameter

α_{12}^S relative volatility between diisopropyl ether (1) and isopropyl alcohol (2) in the presence of the solvent

γ_i activity coefficient

Subscripts

i, j components i and j

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Erratum

Erratum to “Effect of pressure and the capability of 2-methoxyethanol as a solvent in the behaviour of a diisopropyl ether–isopropyl alcohol azeotropic mixture”
[Fluid Phase Equilib. 262 (2007) 271–279]

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The publisher regrets the error printed in Table 3 of this article. The correct table is reprinted below.

Table 3
Vapour pressure parameters

Compound	Eq. ^a	A_i	B_i	C_i	D_i	E_i	Reference
Diisopropyl ether (1)	(2)	41.631	−4668.70	−12.792	6.3693×10^{-4}	1	Daubert and Danner [7]
Isopropyl alcohol (2)	(2)	92.935	−8177.10	−10.031	3.9988×10^{-6}	2	Daubert and Danner [7]
2-Methoxyethanol (3)	(1)	15.05	3499.67	−62.061			This work
	(2)	202.63	−12472.00	−27.385	2.6400×10^{-5}	2	Daubert and Danner [7]
	(3)	7.95	1793.98	236.88			Pick et al. [8]

^a Vapour pressure equations: (1) $\ln P^0$ (kPa) = $A - B/[T(K) + C]$; (2) $\ln P^0$ (Pa) = $A + B/T(K) + C \ln T(K) + D(T(K))^E$; (3) $\log P^0$ (mmHg) = $A - B/[T(^{\circ}\text{C}) + C]$.

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